CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 23: Graphical Representation (Histograms and Ogives)

Previous year board questions, topic-wise: 13 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Mode from a histogram

1. Draw a histogram for the marks of 164 students given below, and use it to find the mode. ICSE 2025, 3 marks

Marks	30–40	40–50	50–60	60–70	70–80
Number of students	10	26	40	54	34

2. A histogram shows the heights of the plants in a school garden. Its rectangles stand on the classes 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and 90–100 (in cm) and have heights 4, 2, 8, 12, 6, 3 and 4. (a) Make the frequency table. (b) State the modal class. (c) Find the mode. (d) How many plants are between 80 cm and 90 cm tall? ICSE 2024, 5 marks

3. The daily wages of 60 workers are given below. Draw a histogram and use it to find the mode. ICSE 2023, 3 marks

Daily wage (₹)	200–300	300–400	400–500	500–600	600–700	700–800
Number of workers	6	10	14	16	10	4

4. Which of these cannot be found graphically for a grouped frequency distribution? (a) median (b) mode (c) quartiles (d) mean ICSE 2023, 1 mark

5. Draw a histogram for the weekly wages below and estimate the mode from it. ICSE 2020, 4 marks

Weekly wages (₹)	3000–4000	4000–5000	5000–6000	6000–7000	7000–8000	8000–9000	9000–10000
Number of people	4	9	18	6	7	2	4

6. The runs scored by 50 batsmen are given below. Draw a histogram and estimate the mode.
ICSE 2018, 4 marks

Runs scored	3000–4000	4000–5000	5000–6000	6000–7000	7000–8000	8000–9000	9000–10000
Number of batsmen	4	18	9	6	7	2	4

2 Reading an ogive

7. An ogive for the monthly salaries of the workers in a factory passes through the points below. Find (a) the total number of workers (b) the median class (c) the lower-quartile class (d) the number of workers earning at least ₹6,000 but less than ₹10,000. ICSE 2025, 4 marks

Salary less than (₹)	2000	4000	6000	8000	10000	12000
Cumulative number of workers	10	20	35	55	70	75

8. The ages of 100 policy holders are given below. Draw an ogive and use it to find (a) the median age (b) the number of policy holders older than 52 years. ICSE 2024, 4 marks

Age (years)	20–25	25–30	30–35	35–40	40–45	45–50	50–55	55–60
Policy holders	2	4	12	20	28	22	8	4
Cumulative frequency	2	6	18	38	66	88	96	100

9. The weights of 60 students are given below. Draw an ogive and use it to find (a) the median (b) the upper quartile (c) the number of students who weigh more than 37 kg. ICSE 2023, 6 marks

Weight (kg)	28–30	30–32	32–34	34–36	36–38	38–40	40–42	42–44
Number of students	2	4	10	13	15	9	5	2

10. The distances thrown by 40 students in a shot-put competition are given below. Draw an ogive and use it to find (a) the median (b) the upper quartile (c) the number of students who threw more than 16.5 m. ICSE 2020, 6 marks

Distance (m)	12–13	13–14	14–15	15–16	16–17	17–18	18–19
Number of students	3	9	12	9	4	2	1

11. The marks of 120 students in a test are given below. Draw an ogive and use it to estimate (a) the median (b) the number of students who failed if the pass mark is 50 (c) the upper quartile.
ICSE 2019, 6 marks

Marks	0–10	10–20	20–30	30–40	40–50	50–60	60–70	70–80	80–90	90–100
Students	5	9	16	22	26	18	11	6	4	3

12. The heights of 60 boys are given below. Draw an ogive and use it to estimate (a) the median (b) the lower quartile (c) the number of boys taller than 158 cm. ICSE 2018, 6 marks

Height (cm)	135–140	140–145	145–150	150–155	155–160	160–165	165–170
Number of boys	4	8	20	14	7	6	1

13. The daily wages of 80 workers are given below. Draw an ogive and use it to estimate (a) the median wage (b) the lower quartile (c) the number of workers who earn more than ₹625 a day.
ICSE 2017, 6 marks

Wages (₹)	400–450	450–500	500–550	550–600	600–650	650–700	700–750
Number of workers	2	6	12	18	24	13	5

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